

Head in the Cloud?

What's on the Horizon for Offline Intelligence

Agenda

Laying the Groundwork - Assistive Technology just the Cloud now?

Case Studies

At Home in the Cloud? Characterizing the Artificial Intelligence layer

Case Studies

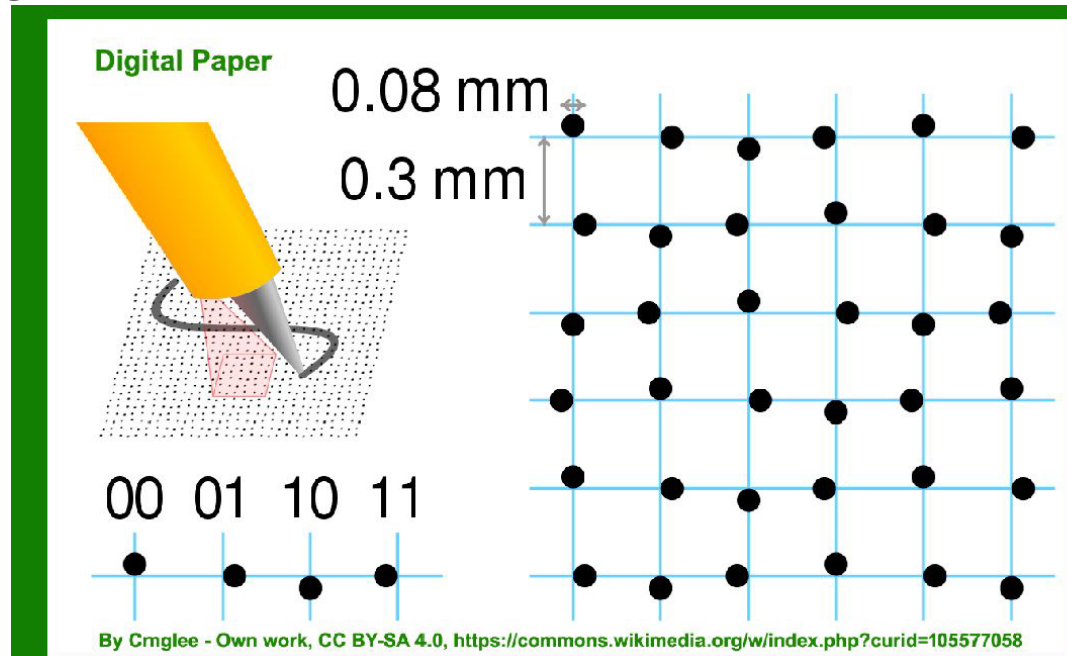
Discussion

Case Study: LiveScribe Smartpen

Complement handwritten notes with on-device audio using specialized pen and note paper

Note paper & ink as the subscription component

Built-in audio recording feature eventually offloaded to accompanying smartphone



Case Study: Voice Dream Reader

Popular Document Text-to-Speech tool (Mac, iOS)

Originally: one-time purchase at \$29.99

Shifted to: monthly subscription of \$9.99 or \$49.99 annually

voice dream

Reader

Blog

[Pricing Update for One-time Purchasers](#)

[Update: April 6th]

Following our recent announcement to transition Voice Dream to a subscription, we received an overwhelming response from thousands in our community. Your feedback, along with the impactful stories shared about Voice Dream being a pivotal part of your daily lives, has led us to reverse this change.

We will continue to provide access to the app's existing features at no additional cost.

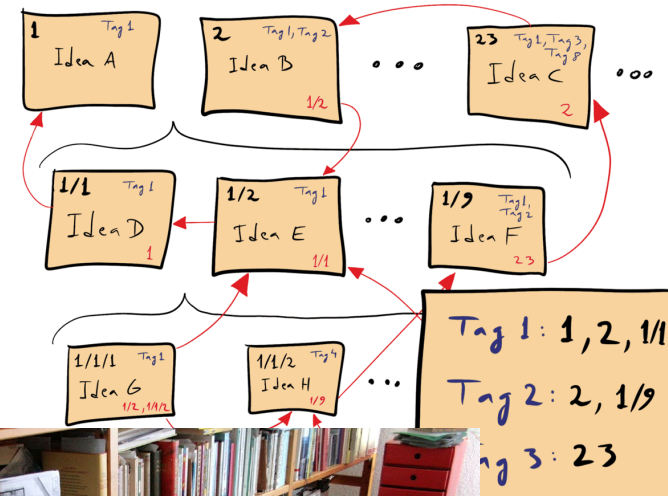
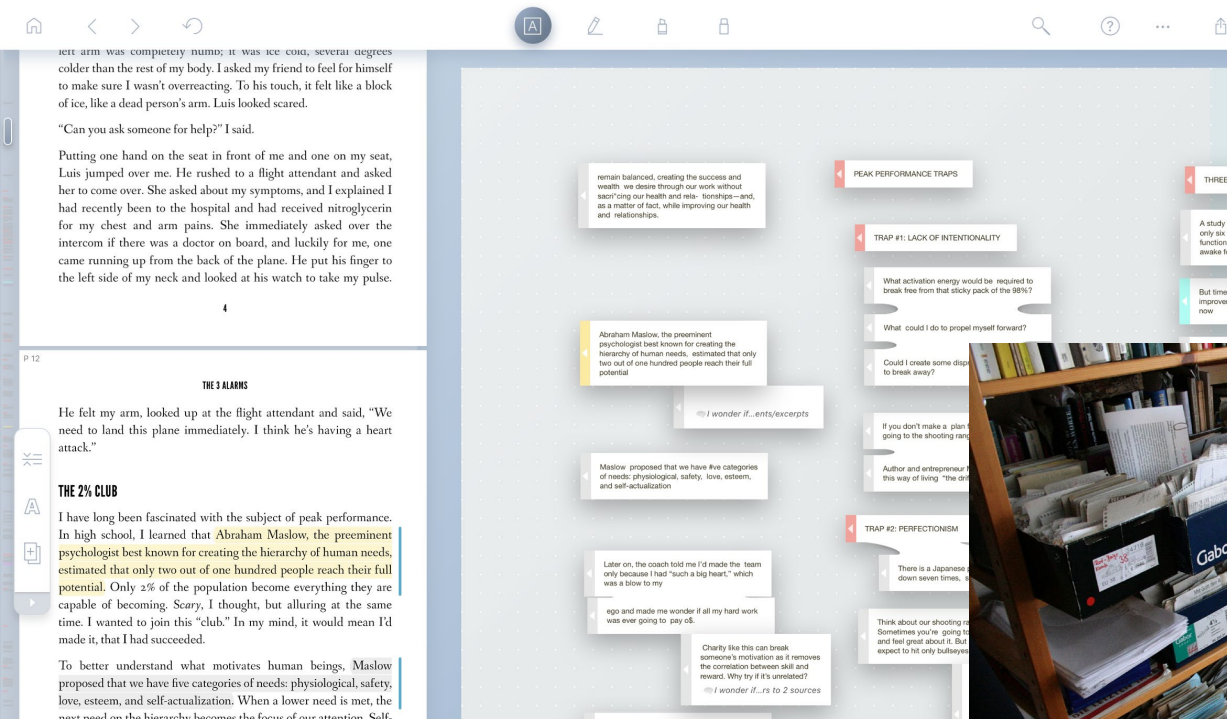
As we continue developing Voice Dream, some new features may be offered as part of a subscription, but the current capabilities will remain free to those who have already purchased Voice Dream.

Case Study: LiquidText

“Zettelkasten” PDF Note-taking tool (Windows, Mac, iPad)

Offers version-specific, one-time purchase pricing

Subscription unlocks cloud features & sync



Flashback: Kurzweil Reading Machine



Cloud-based App Subscription Model

Yay!

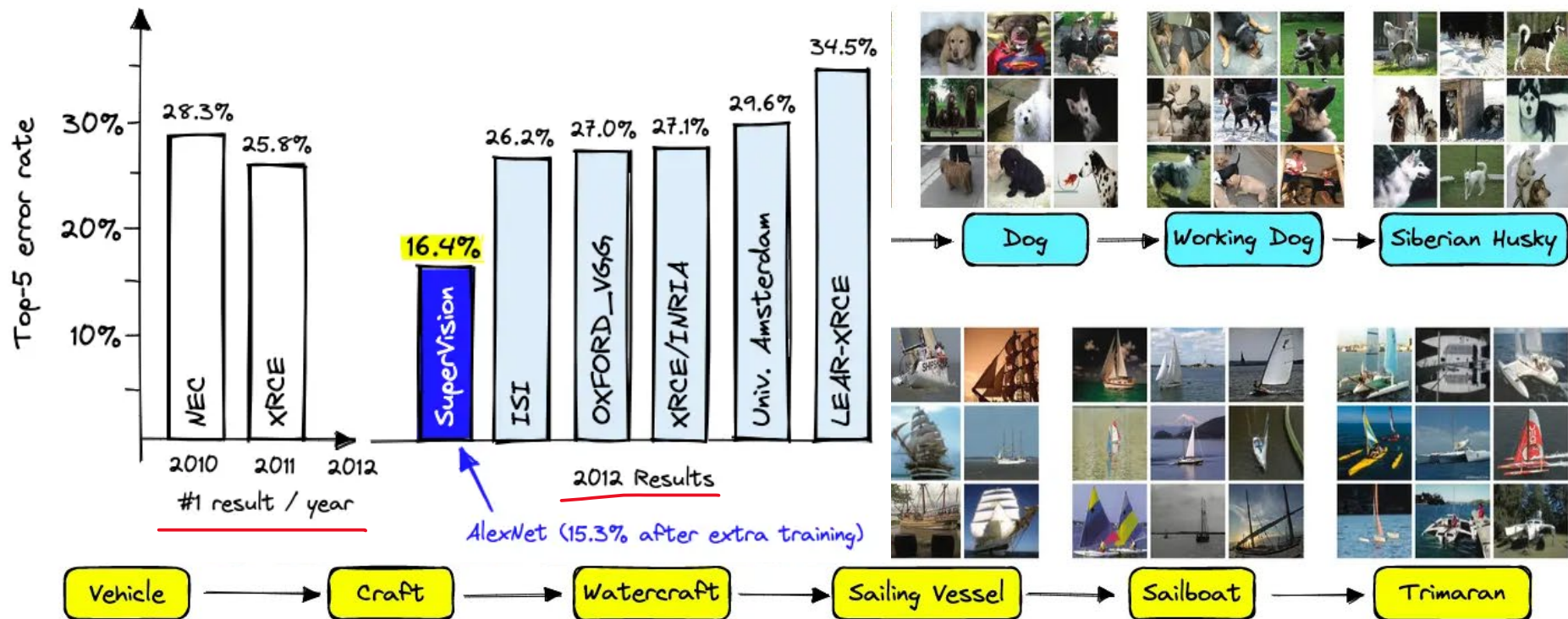
- Bring your own device
- Instant access to latest technology
- Continuous development
- Competition
- Low entry cost

Ugh!

- Recurring, often increasing per-seat fees
- Introductory vs renewal pricing
- “Mission-creep”
- Complicated privacy and accessibility evaluations subject to changes over time

Artificial Intelligence (AI) Primer

ImageNet Database and associated competition



Let's check it out!



Offline slide



Assistive Technology Application

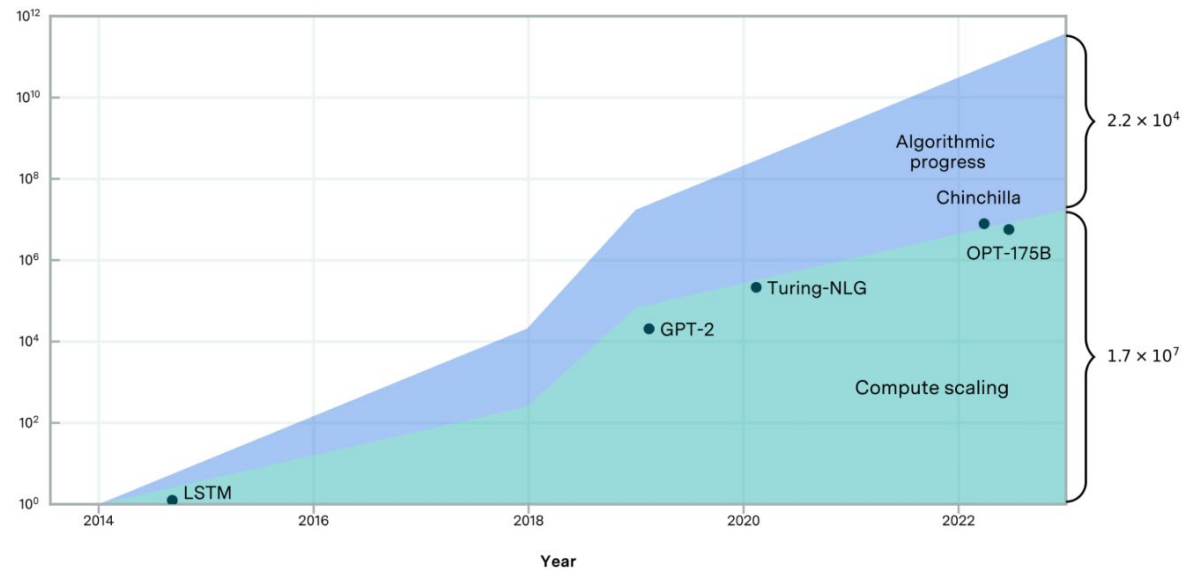
Pattern Recognition and Generation

Not just Bigger and Better

Relative contribution of compute scaling and algorithmic progress

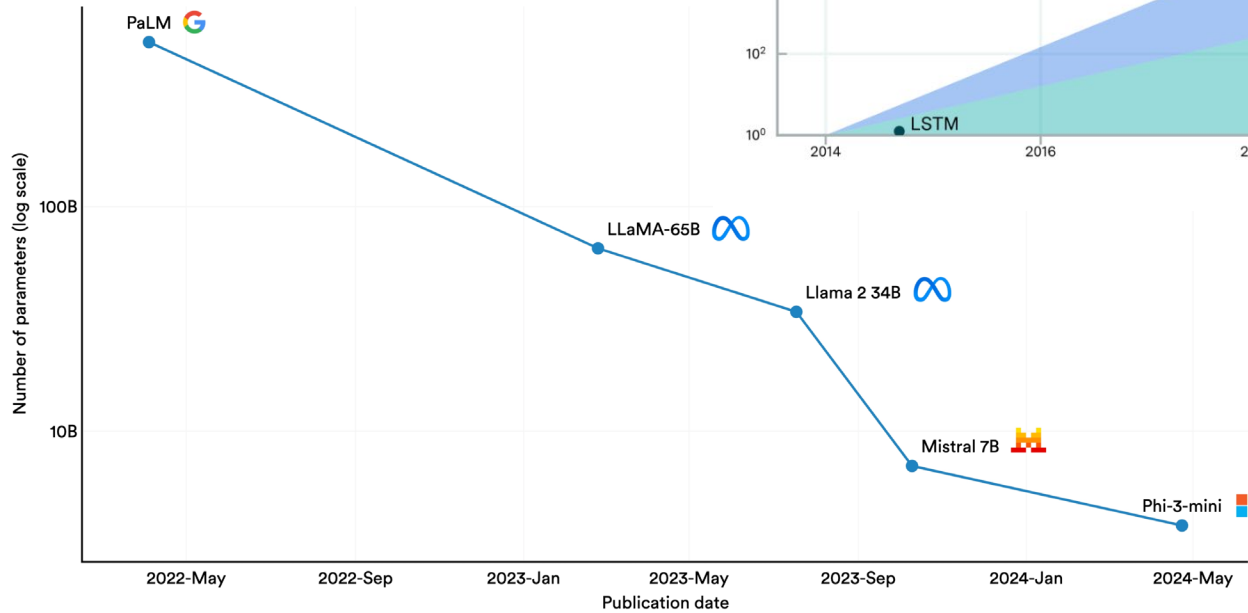
EPOCH AI

Effective compute (Relative to 2014)

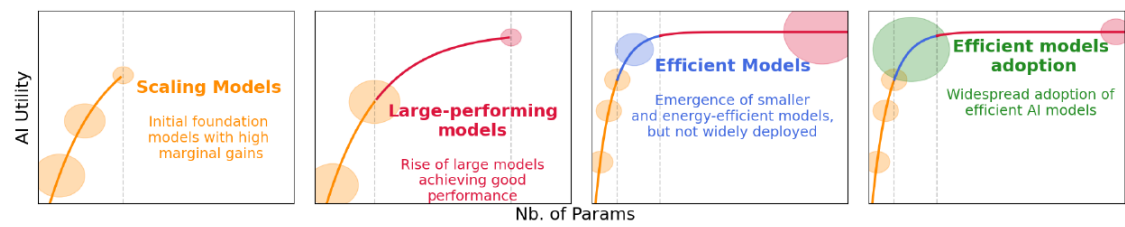


Smallest AI models scoring above 60% on MMLU, 2022–24

Source: Abdin et al., 2024 | Chart: 2025 AI Index report



Small is Sufficient

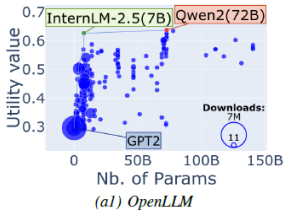


(a) The 4 stages of development of an AI task over time

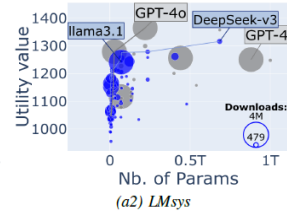
Benchmark Source

- Hugging Face (Blue)
- Papers With Code (Orange)
- Web (Grey)

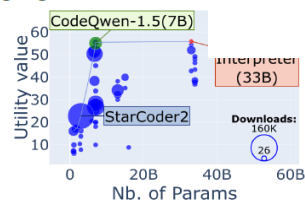
Language



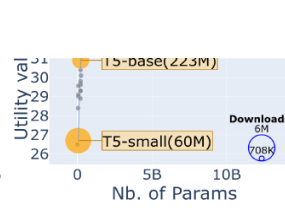
(a1) OpenLLM



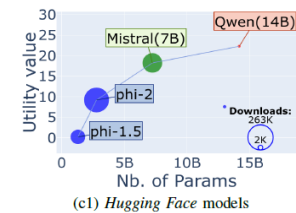
(a2) LMSys



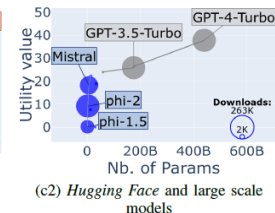
(b) Code Generation



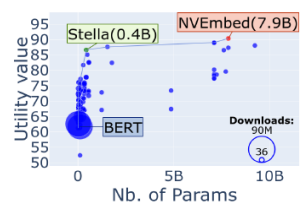
(c) Translation



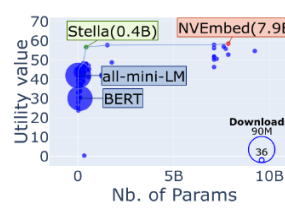
(c1) Hugging Face models



(c2) Hugging Face and large scale models

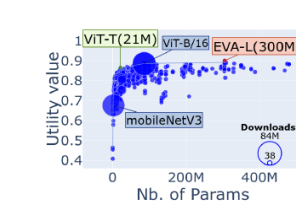


(e) Text Classification

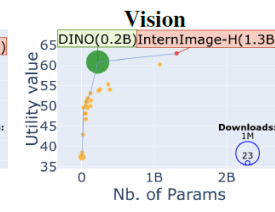


(f) Text Clustering

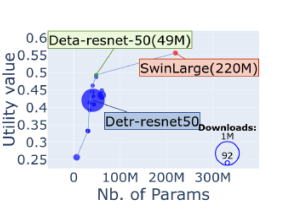
(d) Mathematical Reasoning



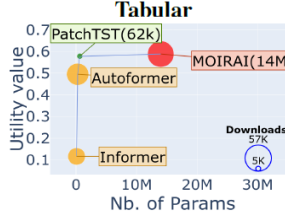
(g) Image Classification



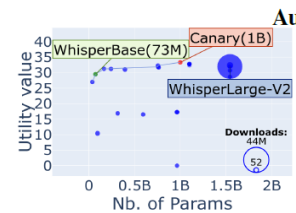
(h) Image Segmentation



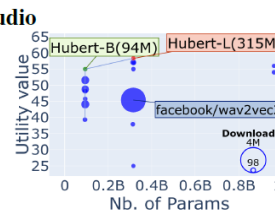
(i) Object Detection



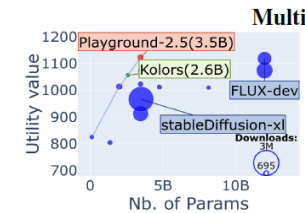
(j) Time Series Forecasting



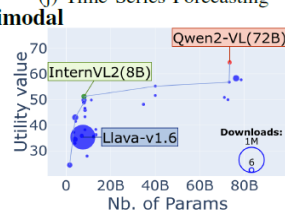
(k) Speech Recognition



(l) Audio Classification



(m) Text to Image



(n) Image-Text to Text

Multimodal

Case Study: Apple Spoken Content

Text-to-speech feature available on Apple devices

Long-standing accessibility feature for offline playback of on-screen text with downloadable voices

30+ languages but premium voices limited to small handful

Once downloaded, a voice becomes available for other text to speech apps

iPhone 12/Apple Silicon: Apple Notes transcription available offline for select languages

Shoutout to Live Speech including offline Personal Voice

Case Study: Google Pixel Recorder App

Speech to text app specific to Google Pixel phones

Works entirely offline using specialized hardware and AI model

First became available packaged with the Pixel 5 (released in 2020)

Includes advanced features such as speaker identification

English-language only

Shoutout to Google AI Edge Gallery for exploring on-device AI capabilities

Case Study: Windows Recall

Chat with snapshots of your screen going back in time (Windows Copilot+ PCs only)

Offline after one-time download of AI components

Concerns about privacy incl. inconsistent filtering of sensitive information

Dependent on user education around local security

Limited to 6 primary languages

Shoutout to Windows 11 Live Captions (plus translation from/into dozens of languages on Copilot+ PCs)

Discussion

Digital technology (cloud-based or not) ...
experiences ongoing development which ...
requires recurring revenue which ...
has to be sold either by subscription (typical for
apps) or by upgrades (typical for hardware, but
also app version-pricing)